Does the model fit the landscape? Shape and scale in Bayesian optimisation and CMA-ES

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Repository: https://github.com/isaacsoong055/DSA4212

Abstract

Bayesian optimisation (BO) and CMA-ES both carry an implicit model of the landscape: CMA-ES a Gaussian search distribution with a covariance (its shape) and a step size (its scale), BO a Gaussian-process surrogate whose kernel metric and lengthscales play the same two roles. We test one explanation in two settings: a method succeeds when its model's shape and scale can match the problem, whatever its family. *Part I (geometry)* asks whether the model can *tilt*: on rotated, ill-conditioned valleys diagonal CMA-ES, the ARD kernel and TuRBO's axis-aligned trust region all fail, while their tilting counterparts do not, once BO's own fitting failures are removed. *Part II (multimodality)* asks whether the model's scale lets a method *find* and then *refine* a narrow distant basin: neither default method finds it; CMA-ES's step size cannot do both jobs at once, and BO succeeds once its lengthscale is capped (1/50 to 44/50 on fresh seeds) while acquisition tuning helps far less. In both parts the decisive BO changes were to the model fit, not the acquisition function. All results are in 2-D and pre-registered predictions are reported whether they held or failed.

1 Shared framing

CMA-ES adapts a covariance matrix and a step size from the ranking of sampled candidates [7]; BO fits a Gaussian-process (GP) surrogate and chooses points with an acquisition function such as expected improvement (EI) [5]. Each carries a model of the landscape. Its *shape* is the orientation and relative stretch it can represent: a full or diagonal covariance for CMA-ES, a full or diagonal (ARD) kernel metric for BO. Its *scale* is the size of region it treats as one feature: CMA-ES's step size σ, BO's lengthscales. Our shared explanation is that a method succeeds when its model's shape and scale can match the problem's, and fails when a restriction or a fitting failure prevents it, whichever family it belongs to.

The two parts test this from different sides of the brief. Part I (Paul Ng) takes the direction *geometry and parameterisation*: conditioning, rotations, changes of units, boundaries and which invariances survive the implementation, plus noisy evaluations. Part II (Isaac Soong) takes *multimodality*: how population size, restarts, step size and acquisition and surrogate choices change the probability of finding a narrow or distant global basin. The parts use different budgets (200 and 300 evaluations), metrics ($\log_{10}$ regret; success rate) and BO implementations (NumPy/SciPy; BoTorch [3]), so their results are never put on common axes. Both follow the same discipline: the budget is counted in objective evaluations, including initial designs and restarts; every comparison reuses the same seeds; predictions were committed to the repository before the runs that tested them; and uncertainty is reported as 95% intervals (bootstrap over instances in Part I, Wilson and Newcombe intervals for success rates in Part II), following COCO practice [9].

2 Part I: can the search tilt? (geometry and parameterisation)

Question and predictions. Near a minimum $f \approx f^* + \frac{1}{2}h^\top Hh$. The Hessian's eigenvalues are the curvatures; their ratio κ is the conditioning (a narrow valley when large); its eigenvectors are the valley's orientation, which a rotation changes without changing κ. A full covariance (CMA-ES) or a full kernel metric $M = LL^\top$ (BO) can align with a rotated valley; a diagonal covariance, the standard ARD kernel and TuRBO's axis-aligned trust region [4] cannot. We predicted that (i) each axis-restricted method pays a penalty under rotation that its tilting counterpart does not, (ii) only when the problem is both ill-conditioned and rotated, and (iii) the tilting version pays a small price when no tilt is needed.

Set-up. 2-D ellipsoids with log-spaced eigenvalues from 1 to κ, the optimum on a radius-2 sphere, aligned or Haar-rotated, 15 instances (COCO-style; seed = instance). CMA-ES follows Hansen's tutorial, with a diagonal ablation. BO is GP-EI (initial design of $2d + 4$, EI over 200 uniform candidates polished by L-BFGS-B, targets log-transformed) with a MAP lengthscale prior on the metric's eigen-lengthscales [10], so that the prior does not favour axis-aligned metrics. Both get the box $[-5, 5]^2$ (BO searches it; CMA-ES starts at its centre with $\sigma_0 = 3$) and 200 evaluations.

CMA-ES. At $\kappa = 1{,}000$, diagonal CMA-ES solves every instance at 0° and 90° and none between 30° and 60°; full CMA-ES solves every angle (Fig. 1). The 90° control only swaps the axes, so the rotation, not the conditioning, causes the failure. At $\kappa = 100$ both succeed everywhere (prediction ii). Full CMA-ES is rotation-invariant in distribution (405 against 408 median evaluations under Haar rotation) and costs up to 21.6% more evaluations on

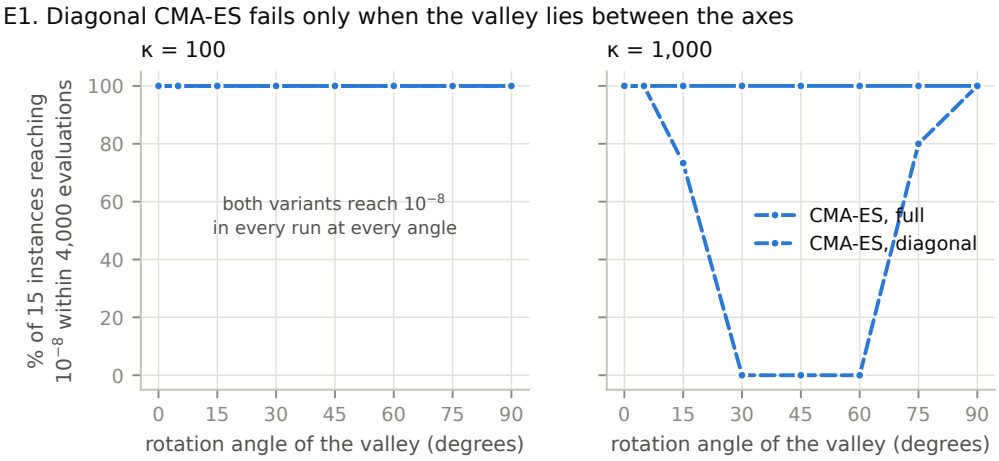


Figure 1. Part I, E1. Success of full (solid) and diagonal (dashed) CMA-ES against the angle of a 2-D valley.

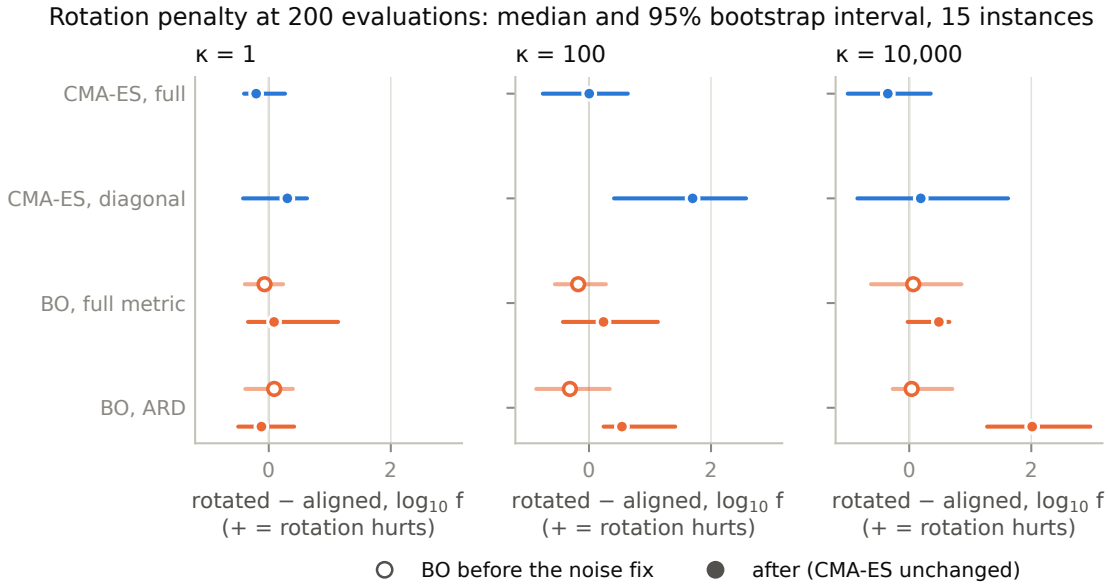


Figure 2. Part I. Rotation penalty at 200 evaluations (median, 95% bootstrap interval, 15 instances). Hollow: BO before the noise fix. The axis-restricted variant pays in both families once BO is no longer stalled.

aligned problems at $d = 32$ (prediction iii). IPOP-style restarts [2] do not repair the diagonal failure (7% against 20% reaching 10^{-5} at rotated $\kappa = 10^4$): a restarted diagonal run still cannot tilt.

BO: a negative result, diagnosed. Reading the fitted kernel directly, ARD's long axis is off by exactly 45° on 45° problems while the full metric is within 0.2°–4°. Yet the first head-to-head showed *no* rotation penalty for BO-ARD (+0.05 [-0.53, +0.63] decades at $\kappa = 100$), while diagonal CMA-ES paid +1.70 [+0.41, +2.57]. BO was stalling everywhere, near $10^{-2.6}$ even on the round bowl. Per-iteration logs ruled out the target transform's floor and the candidate search, and showed the cause: the GP fitted a noise term to a noise-free function and explained its best points away as noise (fitted noise SD 0.2–0.4; posterior mean at the incumbent 2–2.5× the gap to the second-best point above its observed value), so EI against the observed best underflowed below 10^{-12} in 97–100% of late iterations. Fixing the noise variance at 10^{-6} lifted BO by −1.17 [-1.71, -0.74] decades; LogEI [1], which fixes the underflow but not its cause, changed nothing (+0.01 [-0.79, +0.47]).

The penalty, once BO works. Re-running the 180 BO runs with the fix, paired by instance, the negative result reversed (Fig. 2): BO-ARD's rotation penalty became +2.01 [+1.28, +2.97] decades at $\kappa = 10^4$ and BO-full's stayed non-significant. Table 1 repeats the key comparison with a second kernel (Matérn-5/2), a second acquisition search (200 extra candidates around the incumbent) and a TuRBO-style trust region whose box is either axis-aligned with ARD-scaled sides, as published, or rotated to the full metric: in every variant the axis-restricted version ends significantly behind on rotated problems and level on aligned ones. With the trust region, BO also beats full CMA-ES at 200 evaluations on aligned $\kappa = 100$ (−1.36 [-2.02, -0.85]) and rotated $\kappa = 10^4$ (−3.32 [-4.24, -1.82]); with plain EI, CMA-ES wins by 200 evaluations at $\kappa \leq 100$ (1.2–2.7 decades).

Beyond the quadratic. A quadratic has the same orientation everywhere; Rosenbrock (COCO f8/f9 [8]) bends,

BO variant (2-D, 15 instances, 200 evaluations)	rotated $\kappa = 10^4$	aligned $\kappa = 10^4$
EI, squared exponential, noise fixed	$+1.40$ [$+0.68, +1.69$]	-0.13 [-1.15, $+0.19$]
EI, Matérn-5/2	$+1.06$ [$+0.59, +1.43$]	-0.13 [-0.70, $+0.54$]
EI + local candidates	$+1.28$ [$+0.12, +2.36$]	-2.96 [-3.44, $+2.61$]
Trust region, ARD box vs rotated box	$+2.92$ [$+1.27, +3.79$]	-0.39 [-3.11, $+4.65$]

Table 1. Part I. Axis-restricted minus tilting version, median $\log_{10} f$ at 200 evaluations, paired by instance (+ = axis-restricted worse). Wide aligned intervals come from the boundary trap (text).

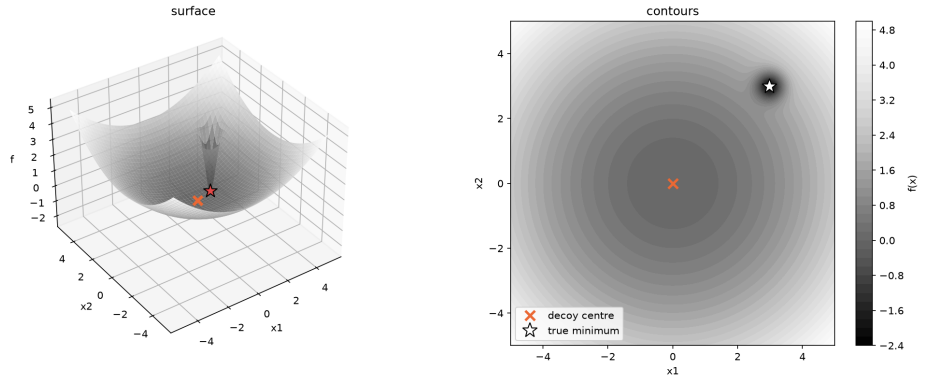


Figure 3. Part II. Primary landscape: a decoy bowl centred at the origin and a narrow global well at $(3, 3)$, $w = 0.3$.

so its local orientation changes along the path, and even "aligned" f8 is tilted 63.5° near its optimum. With 4,000 evaluations full CMA-ES reaches 10^{-5} in every run (median about 500 evaluations), diagonal in 0% (f8) and 27% (f9): what matters is the local orientation. For BO (300 evaluations) the trust region beats plain BO by 3.5–3.7 decades and its tilted box beats the axis-aligned one by 2.0–2.5 decades, even on f8. We predicted that fitting the GP only to points inside the region would follow the bend better; it was 1.8–2.0 decades *worse*, and worse still (2.7) on a quadratic control where local and global shape coincide, so the loss is discarded data, not geometry.

Invariances, units, boundaries. CMA-ES runs were bit-identical to the baseline under every monotone transformation of f (120/120); BO is not rank-invariant, and cubing f costs it 1.7 decades through its relative target floor. A change of units hurts every method with default set-ups (CMA-ES about 3.9 decades, BO 1.3–1.4); mapping CMA-ES's initial covariance as Hansen requires ($C_0 = S^{-2}$) restores invariance. BO over-samples the box edge [11]: it is 1.62 / 0.73 decades worse when the optimum lies 0.1 inside an edge and better when it lies on it. Combined with search around the incumbent this produces a *boundary trap*: 4–8 of 15 runs per aligned ill-conditioned cell stall on the valley floor at the box edge. In those runs about two random candidates per iteration are truly better, yet the GP gives them EI near 10^{-34}: a model failure. Reflecting out-of-box candidates instead of clipping them removes the trap for the trust region (6 and 4 stuck runs to 0) but not for local candidates.

Noise. On the $\kappa = 100$ quadratic with additive noise $\sigma \in \{0.01, 0.1, 1\}$ and every recommendation scored on the noise-free f, BO's posterior-mean pick beats its lowest reading by up to 1.08 decades, CMA-ES's mean helps only with a larger population ($\lambda = 24$), and spending evaluations on repeats or a final re-evaluation reserve did not pay at 200 evaluations. With each family's best settings the two are level at every noise level. The one anomaly, a worse model-based pick with replicates, is again the noise-fitting GP: a correctly specified GP on the raw values never does worse than the lowest reading.

3 Part II: finding and refining a narrow basin (multimodality)

Question and predictions. At a fixed budget of $N = 300$ evaluations, how do population size, restarts, step size and acquisition and surrogate choices change the probability of finding a narrow, distant global basin, and why? Predictions H1–H6 and M1–M3 were committed before any optimiser result was inspected; BO1–BO13 and CMA1–CMA6 were added later, each before the experiment that tested it, except BO1 (post hoc, labelled as such).

Set-up. The primary landscape is a broad decoy bowl with a narrow, deep Gaussian well, $f(x) = \|x - c\|^2/s - A\exp(-\|x - x_g\|^2/2w^2)$ with $c = 0$, $s = 10$, $A = 4$, width $w = 0.3$, the well at $(3, 3)$ (Fig. 3), on $[-5, 5]^2$. A run succeeds when its regret falls to $\varepsilon = 0.05$, a threshold chosen from random sampling alone. CMA-ES is the `cma` library (default $\lambda = 6$, $\sigma_0 = 2$, start $x_0 \sim U[-4, 4]^2$); BO is BoTorch GP-EI (Matérn-5/2, 10 initial points, refit every 5 evaluations, lengthscale upper bound 20, fixed noise 10^{-6}). Each variant changes one setting and uses seeds 1000–1049; success rates carry Wilson intervals and differences Newcombe intervals.

Baselines (Exp 0). Neither default finds the well: CMA-ES 0/50, BO 5/50, random sampling 2/50, with

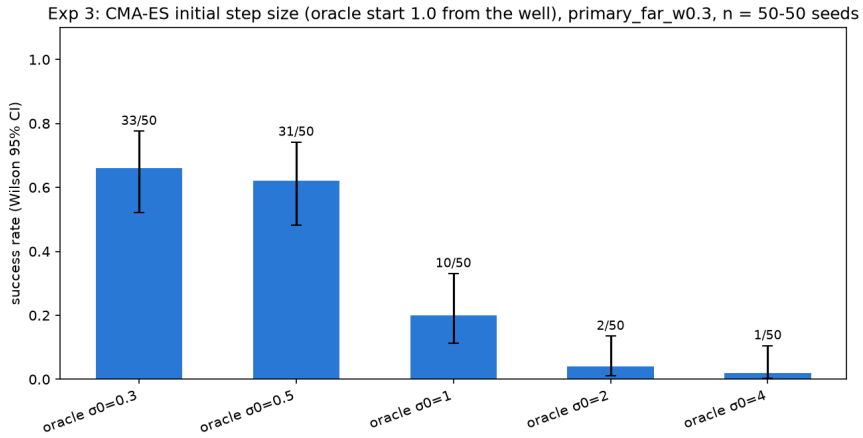


Figure 4. Part II, Exp 3. CMA-ES success against initial step size from a start 1.0 from the well (a diagnostic of refinement, not a method comparison).

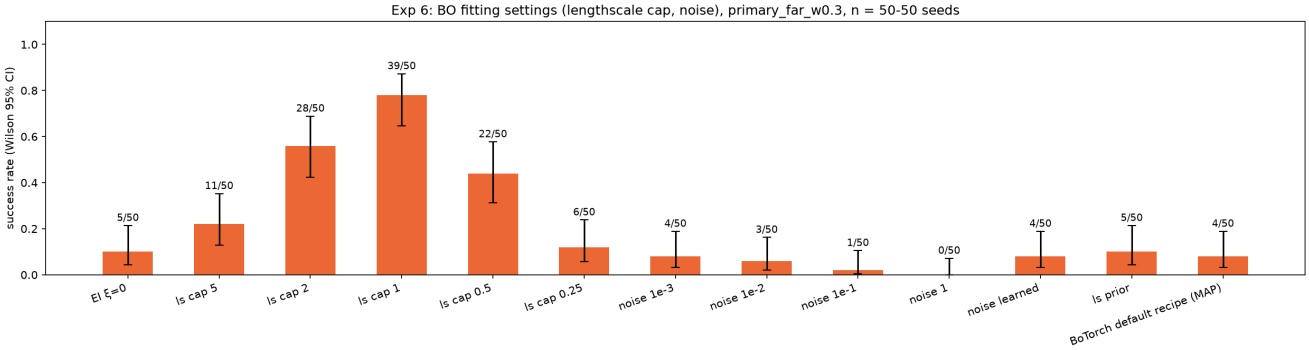


Figure 5. Part II, Exp 6. BO fitting settings changed one at a time: lengthscale cap and noise level (BoTorch, $N = 300$).

pairwise-overlapping intervals. CMA-ES's σ falls below 0.2 by generation 19 (median) and from evaluation 100 on at least 97.5% of its samples lie within 0.5 of the decoy centre (H1 supported). BO's lengthscale runs to its upper bound in 42 of 50 runs, none of which succeeds, and in the 45 failures no sample ever enters the well. Both fail at *discovery*, not refinement: every BO run that sampled the well succeeded.

CMA-ES: population, restarts and step size (Exps 1–3). A larger population raises the number of runs that touch the well (1, 7 and 13 of 50 for $\lambda = 6, 12, 24$) but success stays 0/50 at every λ, also with five times the budget (H2 contradicted). IPOP and BIPOP [2, 6] are identical at this budget, because only one restart fits, and also give 0/50 (H3 contradicted on success). The step size explains why (Fig. 4): from a start 1.0 from the well, $\sigma_0 = 0.3$ succeeds in 33/50 runs and the default $\sigma_0 = 2$ in 2/50 (+0.62 [+0.45, +0.74]), while from a random start no σ_0 exceeds 1/50. A mean trace shows the mechanism: with $\sigma_0 = 2$, a single well sample is outvoted by the other selected points on the bowl, which drag the mean back to the decoy. Finding the well needs a large σ and refining it a small one, and the default has to choose.

BO: acquisition against model fit (Exps 4–7). Exploration in EI helps a little: $\xi = 0.1$ reaches 17/50 (+0.24 [+0.08, +0.39] over $\xi = 0$), and larger ξ touches the well more often but refines it less (17/45 to 3/42 given a touch). LCB (β from 0.5 to 32), Thompson sampling and BoTorch's default MAP recipe all stay at 3–4/50. The model fit matters far more (Fig. 5): capping the lengthscale at 1 gives 39/50 (+0.68 [+0.51, +0.79]), because the default fit runs to the bound and the search stays near the bowl centre, while a cap makes it sweep the box (about 100 of 100 grid cells visited, against 10). The cap has an optimum: 0.25 drops back to 6/50. Changing the kernel at the default cap changes nothing; at cap 1 RBF matches Matérn-5/2 (39/50) but the rougher Matérn-3/2 and 1/2 collapse (6 and 3/50) by failing to discover the well.

Width, control and validation (Exps 8–10). Across well widths (Fig. 6) the capped BO, evaluated on fresh seeds 2000–2049 to avoid selection bias, succeeds in 44/50 at $w = 0.3$ and 50/50 at $w = 0.6$ and 1.0, against 1, 10 and 39/50 for default BO; CMA-ES needs a wide well (35/50 at $w = 1.0$, 0/50 at $w = 0.3$), and the best CMA-ES variant ($\sigma_0 = 1$) gives no gain at any width. On a near well, default CMA-ES beats default BO (30 against 13/50 at $w = 0.3$). The capped BO has a cost on a unimodal sphere (regret 8.9×10^{-4} against 1.2×10^{-6} at evaluation 100), and a larger λ worsens CMA-ES there (H5 supported). On Rastrigin and Schwefel the cap's gain is small and uncertain, and BO is far stronger than CMA-ES on Schwefel (16/20 default, 20/20 capped, against at most 8/50); whether λ helps more on Rastrigin than on Schwefel (H6) could not be distinguished.

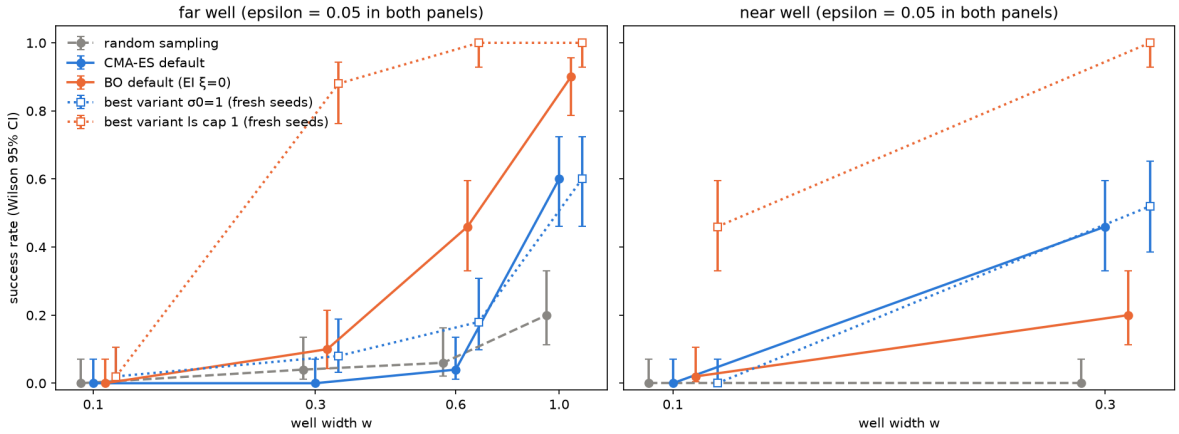


Figure 6. Part II, Exp 8. Success against well width for the defaults and the best variants (fresh seeds), far and near well.

4 Synthesis: what the two parts establish together

Shape and scale both decide success, within each family. Part I shows that a model restricted in *shape* fails on tilted valleys in both families, and Part II that a model with the wrong *scale* fails to find or to refine a narrow basin. In neither part does the family alone predict the outcome: diagonal and full CMA-ES differ by up to 100 percentage points on the same rotated problem, and capped and default BO by 43 successes out of 50 on the same landscape (44 against 1 on fresh seeds).

BO's decisive fixes were to the model fit, not the acquisition. Independently, both parts found BO failing because its GP was mis-fitted, and both found acquisition changes nearly powerless: in Part I the fitted noise absorbed the best points (fixing it gained over a decade; LogEI nothing), in Part II the lengthscale ran to its bound (capping it took success from 5 to 39/50; ξ, LCB and Thompson sampling at most 17/50). In both, BO's results depend on implementation choices that published defaults leave open.

CMA-ES adapts shape well but must commit to a scale. Its covariance follows a rotated valley reliably (Part I), but one step size cannot both find a distant basin and refine a narrow one (Part II); population size and restarts, which change only how much is sampled, did not move success on the primary landscape.

Trade-offs. BO leads early in both parts and costs roughly 10^5 times more computation per run; settings that help on the hard landscape cost on easy ones (the cap on the sphere, tilting on aligned problems); and BO's boundary preference appears in both parts (edge sampling under large ξ; the boundary trap).

Limitations. All headline results are 2-D, so dimension effects are inferred (for CMA-ES) rather than measured for BO. Each part rests on few landscapes (ellipsoids and Rosenbrock; one primary landscape plus two validation functions) and 15–50 seeds, which leaves some differences unresolved. BO results reproduce in distribution only: long BO runs are not bit-repeatable even with one BLAS thread. Some follow-ups were added after seeing results and are labelled as such; untested mechanisms are stated as hypotheses.

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